

INTRODUCTION TO TIME SERIES DATA ANALYSIS AND NEURAL NETWORK

Lab Objectives

By the end of this lab, participants should be able to understand the time series data analyzing, visualizing, manipulating, classification and forecasting in Orange. Besides, basic concept of neural network (NN) will also introduce to the time series data classification problem.

Prior to this lab, participants are required to have basic knowledge of neural network architecture and time series data analysis. The objectives of this lab exercises are to ensure participants familiarise to manipulate time series data under “Orange” software. This lab sheet is structured as follows:

i. Working with Time Series Data

PART A: Install Add-ons in Orange

PART B: Download and load dataset

PART C: Pre-processing and visualisation

PART D: Data forecasting and model evaluation using Classical / Statistical Models

PART E: Case study

ii. Working with Neural Network

PART A: Introduction to Orange NN

PART B: Classification using NN

PART C: Understanding time series data for classification/prediction

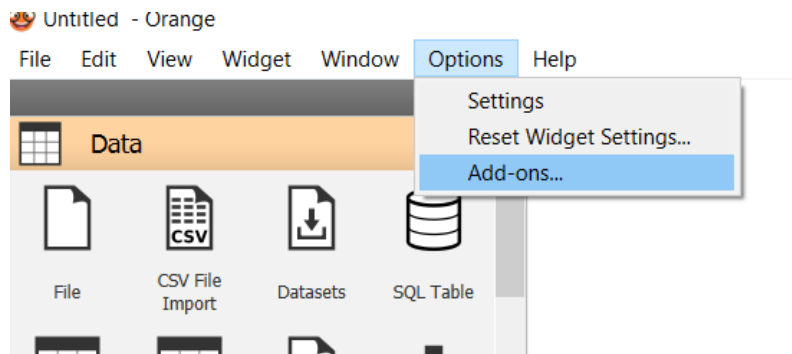
PART D: Time series data classification using NN

PART E: Time series data prediction using Machine Learning

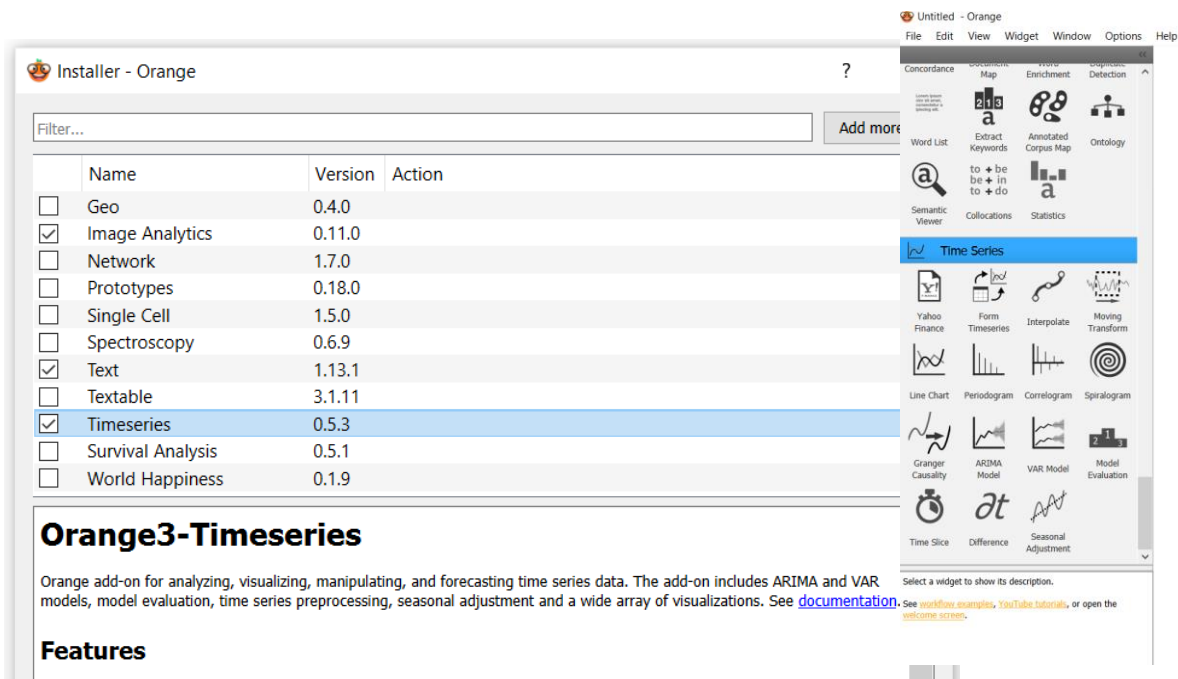
Working with Time Series Data

PART A: Install Add-ons in Orange

The standard Orange software does not come with the time-series data manipulation. Hence, we need to install the additional add-ons to the software. Open the Orange, go to the upper tab and select Options → Add-ons...



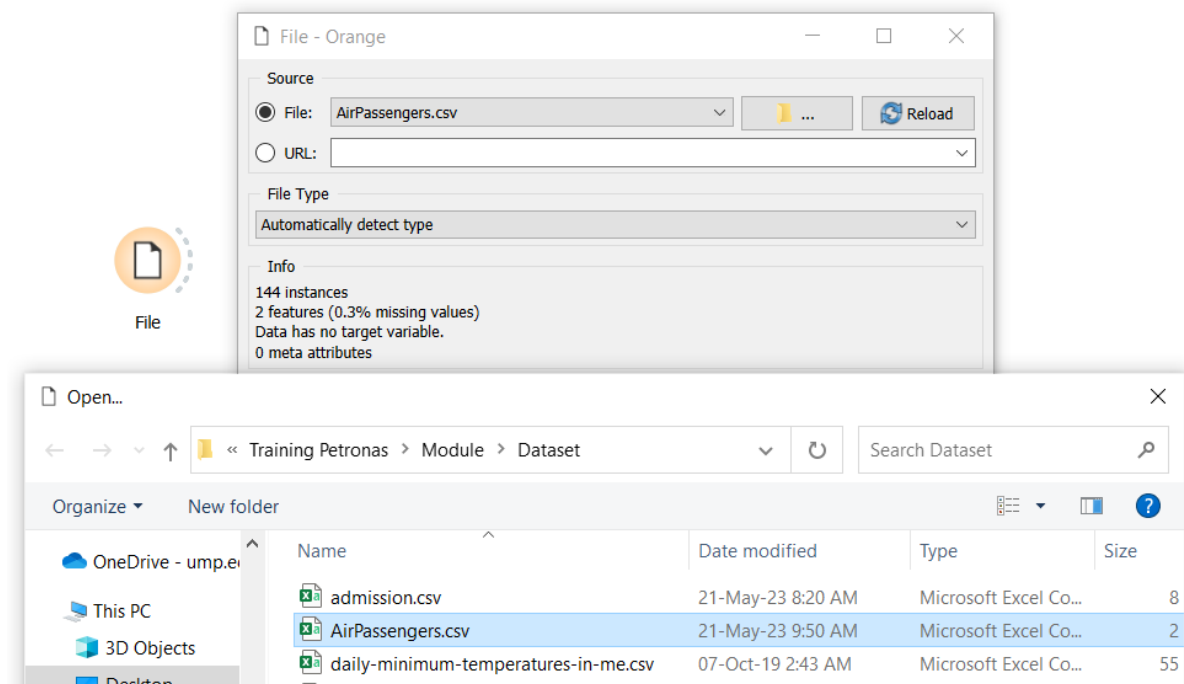
Prior to that, you need to start the software as a system administrator to allow the installation process. In the Installer menu, select “Timeseries”.



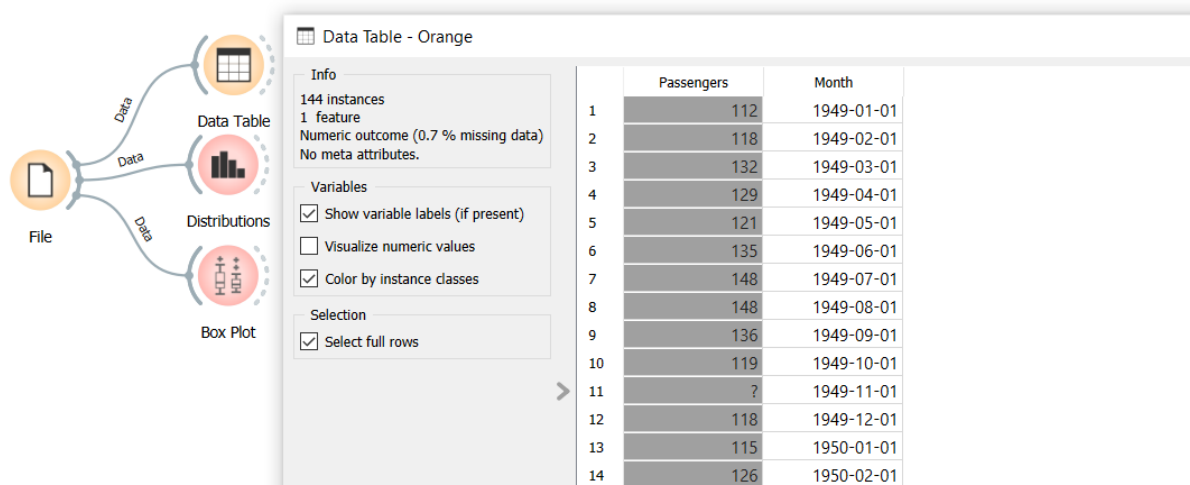
You need to wait until the installation finish. The installation process will take sometimes. After installation finish, restart Orange. You will find the Time Series menu on the left panel.

PART B: Download and load dataset

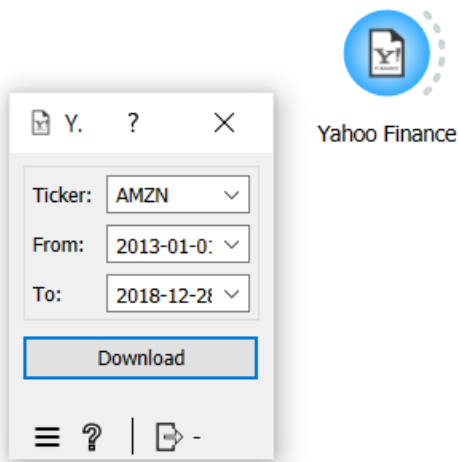
Let's start working with the Orange time series data. We need a dataset in order to play around with time series. In this example, we use "AirPassengers.csv" data that being provided to you. Choose File, browse the folder of the data, and load the dataset.



You can also use a simple data visualization to see the loaded data. There are 144 instances with 0.7% missing data. You can see the missing data is in line 11.



Instead of having own data, you may also generate time series data from Yahoo Finance stock market data.



In this widget, you can choose:

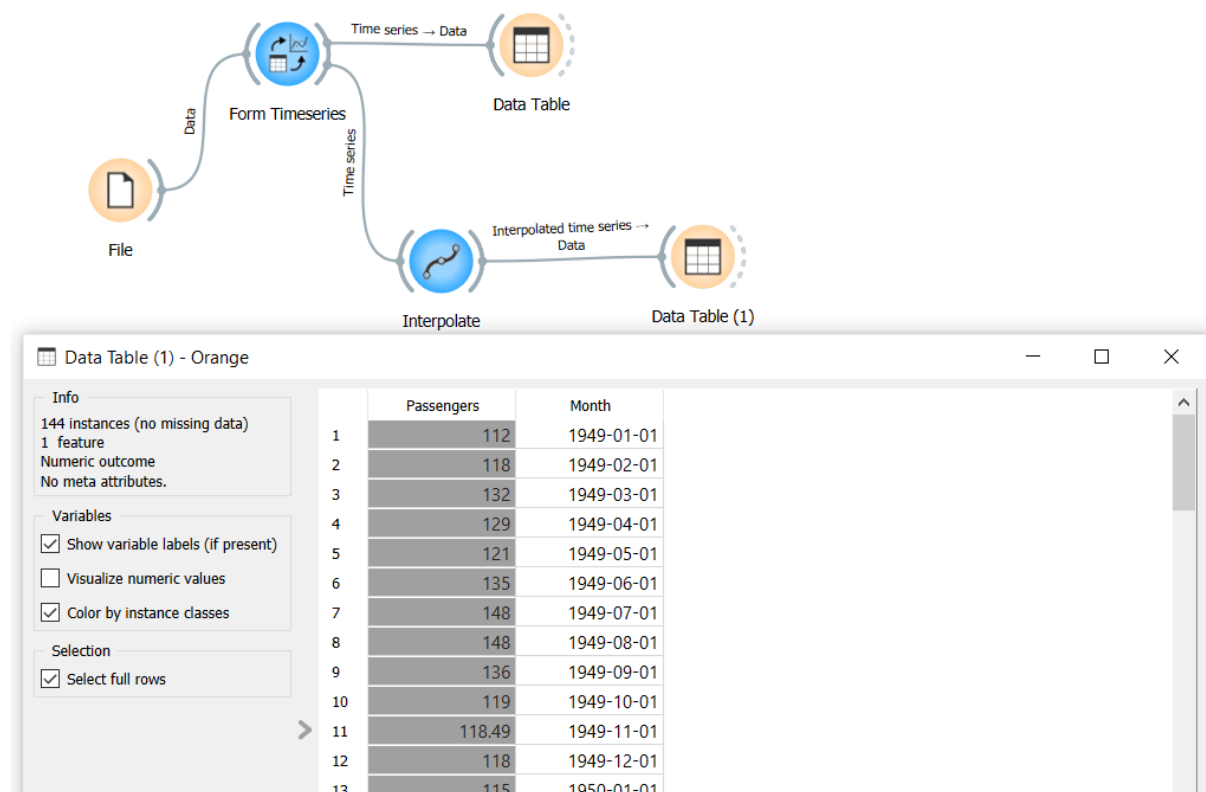
- Ticker (e.g. GOOG) or index (e.g. DJI) symbol you are interested in.
- Date range you are interested in by choosing From until To.

PART C: Pre-processing and Visualisation

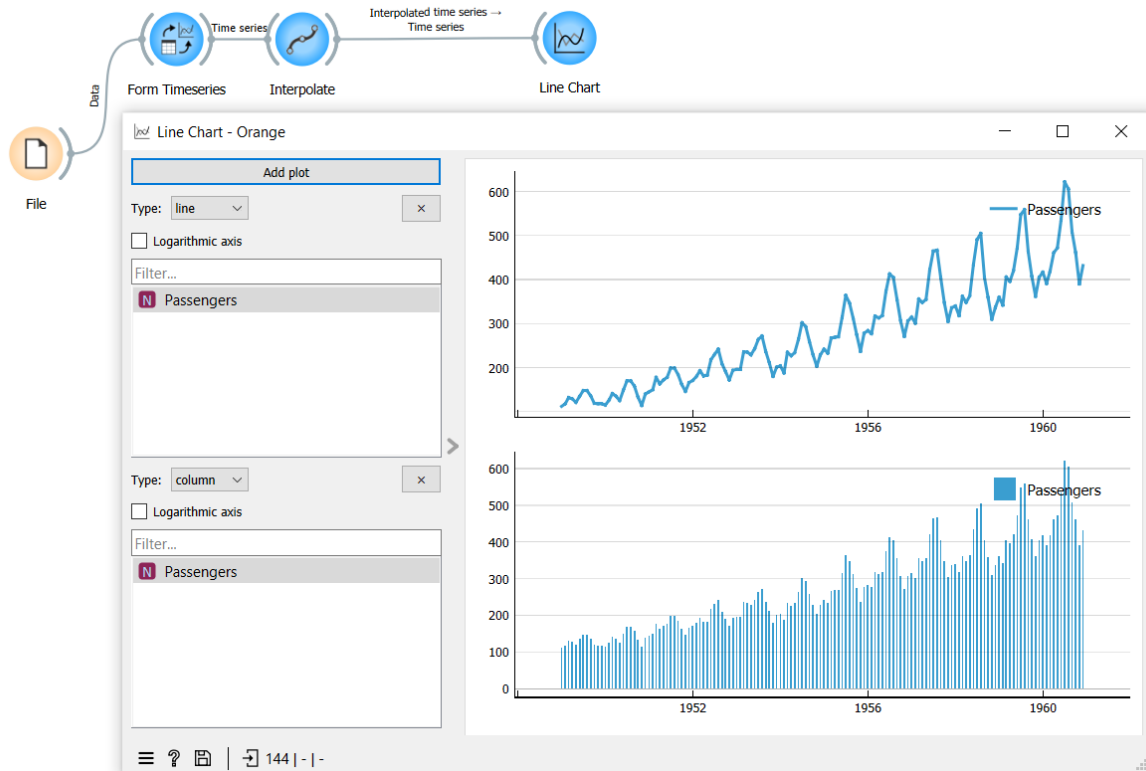
You need to apply the conversion into time series by using “Form Timeseries” widget. In order to induce missing values in the time series by “Interpolation” widget. From the tabulated data table, we can see data in Line 11 has been induced with 118.49. Most time series algorithms assume, you don’t have any missing values in your data. In this widget, you can choose the interpolation method to impute the missing values with. By default, it’s linear interpolation (fast and reasonable default).

Interpolation type. You can select one of linear, cubic spline, nearest, or mean interpolation:

- Linear interpolation replaces missing values with linearly-spaced values between the two nearest defined data points.
- Spline interpolation fits a cubic polynomial to the points around the missing values. This is a painfully slow method that usually gives best results.
- Nearest interpolation replaces missing values with the previous defined value.
- Mean interpolation replaces missing values with the series’ mean.



After the induction process, connect the “Interpolate” to the “Line Chart” widget. You can visualize the time series in this widget. Note that the line option will display the data as a connected line. In case of non-existent dates, the widget will show them, then the line will be drawn over to connect the last known value with the next one. In case of missing values, the widget will not draw the line in the given place, making the chart disconnected. To better see the missing values, we recommend using the column option.



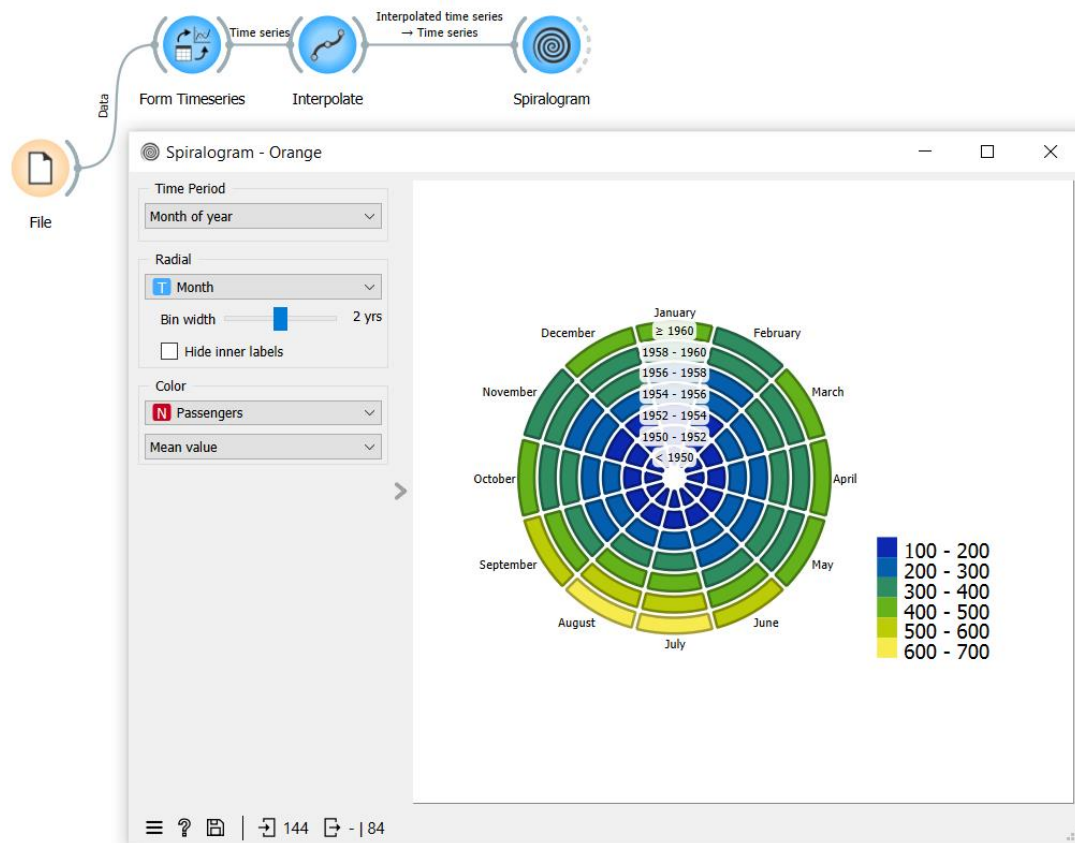
The setup available in this widget:

- Add a new line chart below the current charts. The widget can display up to 5 parallel charts.
- Set the type of chart. Options are line, step line, column, area. X button removes the associated chart.
- Switch between linear and logarithmic y axis.
- Select the time series to display (select multiple series using the Ctrl/Cmd key). Filter enables searching for the desired variable from the list by its name.

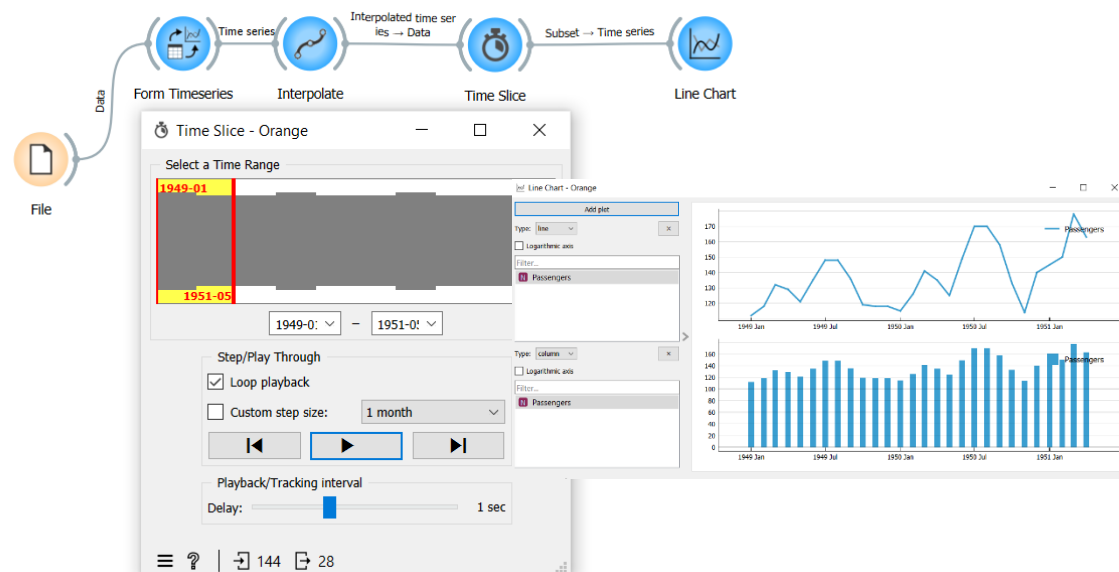
Spiralogram is intended for visualizing time series and for comparing attribute values by categorical variables. Instances can be selected from the plot and sent downstream.

- Units on the circumference. Options are month of year (12 units), day of year (365 units), day of month (~30 units), day of week (Mon-Sun, 7 units), hour of day (24 units) and all the variables from the data.
- Unit of the vertical axis. Hide inner labels removes labels on the vertical axis.
- Color of each spiralogram section. Default is Show instance count. If an attribute from the data is selected, aggregation methods become available. The options are: mean value, sum, product, minimum, maximum, span, median, mode, standard deviation, variance, harmonic mean, geometric mean, non-zero count, and defined count.

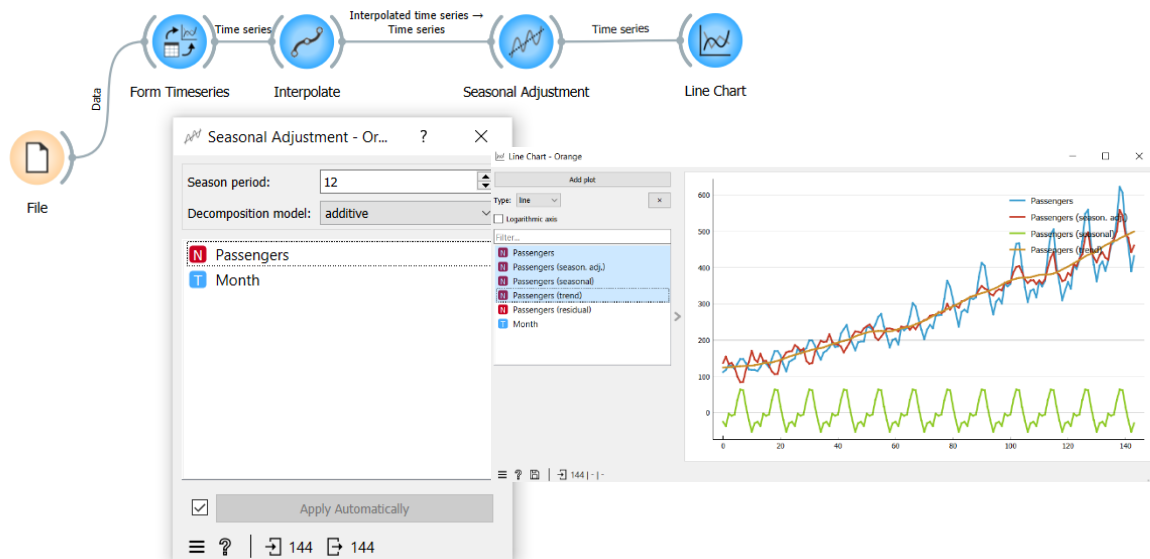
Introduction to Time Series Data Analysis and Neural Network – Lab



Time slice is a subset selection widget designed specifically for time series and for interactive visualizations. It enables selecting a subset of the data by date and/or hour. Moreover, it can output data from a sliding window with options for step size and speed of the output change.



- Visual representation of time series with the selected time slice. Click and drag the red lines to adjust the time window or click and drag the yellow frame to move it around. Alternatively, set the to and from dates below to output the desired subset.
- If Loop playback is selected the data will ‘replay’ continuously. Custom step size defines how the time slice move. If it is set to, say, 1 day, the window will output $n + 1$ day once it moves. Without the custom step size defined, the slice will move to the next frame of the same size without any overlap. Press play to bring the sliding window and stop to stop it. Backwards and forwards buttons move the slice by the specified step size.
- Set the speed of the sliding window.



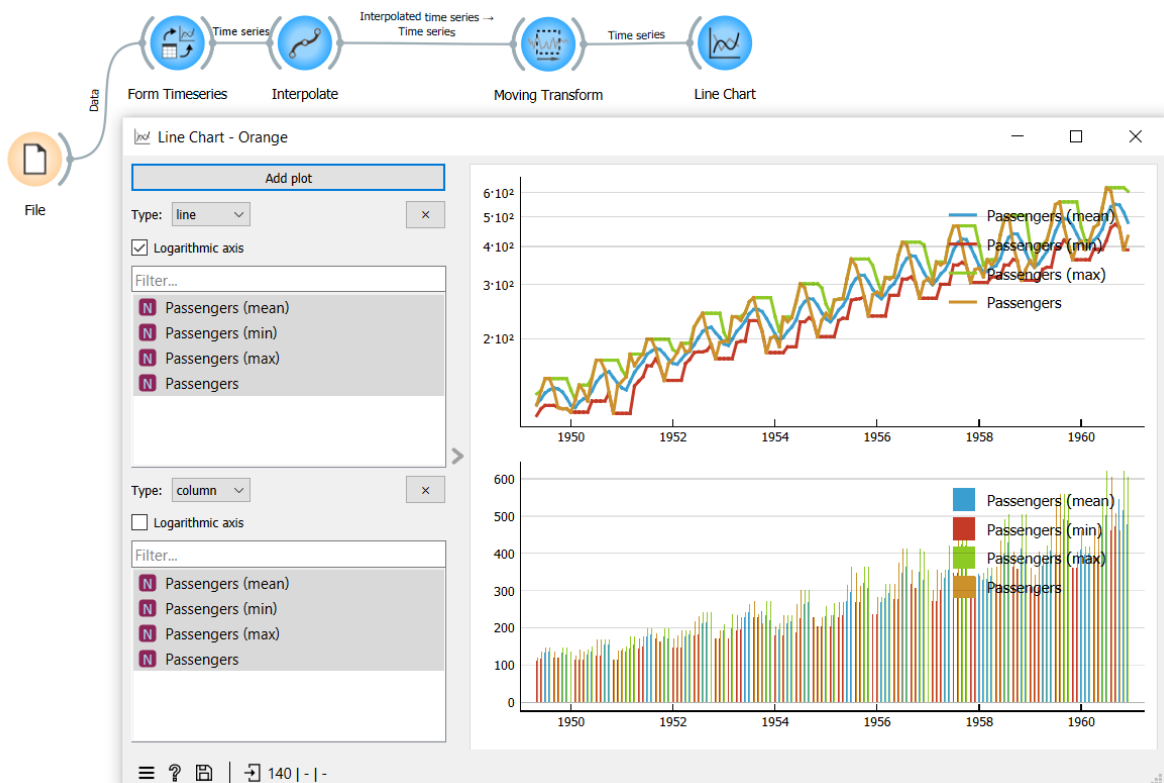
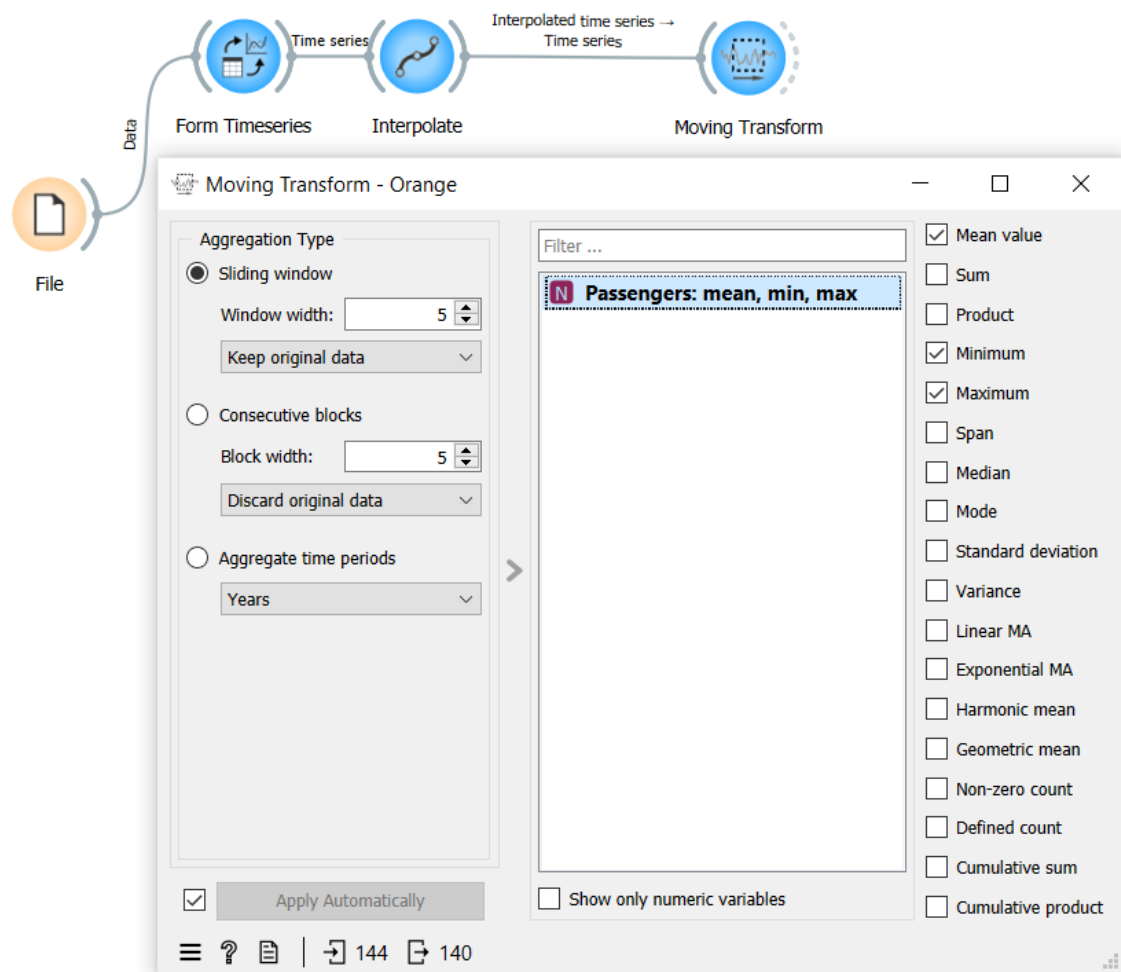
Decompose the time series into seasonal, trend, and residual components.

- Length of the season in periods (e.g. 12 for monthly data).
- Time series decomposition model, additive or multiplicative.
- The series to seasonally adjust.

The “Moving Transform” widget is to compute aggregations over a sliding window, consecutive blocks or time periods of time series.

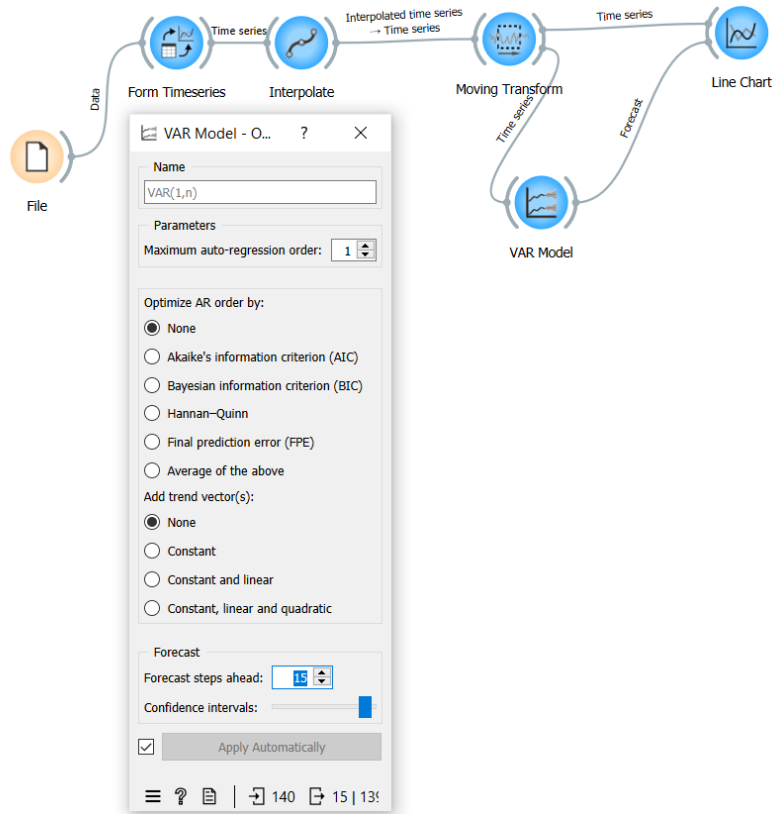
We passed the data to Moving Transform. To get a 5-day moving average, we used a rolling window with mean, min and max aggregation. In Line/Column Chart, we can see the initial (blue) values of High variable are now changed to mean, min, max aggregation.

Introduction to Time Series Data Analysis and Neural Network – Lab



PART D: Data forecasting and model evaluation using Classical / Statistical Models

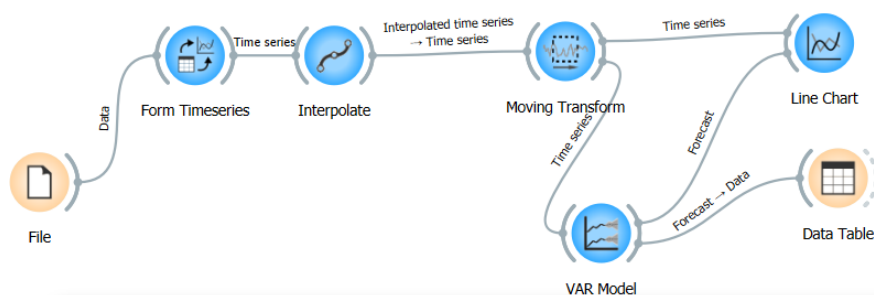
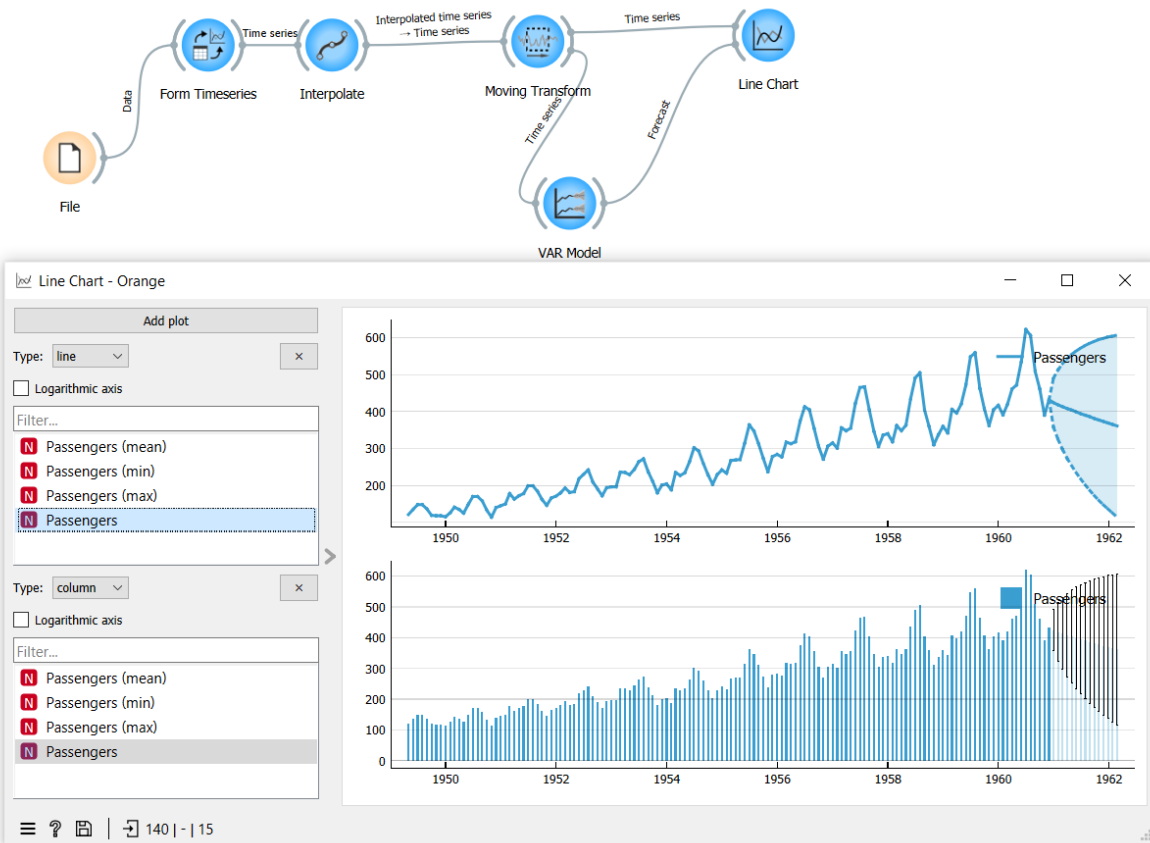
Model the time series using vector autoregression (VAR) model. Using this widget, you can model the time series using VAR model



- Model's name. By default, the name is derived from the model and its parameters.
- Desired model order (number of parameters).
- If other than None, optimize the number of model parameters (up to the value selected in (2)) with the selected information criterion (one of: AIC, BIC, HQIC, FPE, or a mix thereof).
- Choose this option to add additional “trend” columns to the data:
 - Constant: a single column of ones is added
 - Constant and linear: a column of ones and a column of linearly increasing numbers are added
 - Constant, linear and quadratic: an additional column of quadratics is added
- Number of forecast steps the model should output, along with the desired confidence intervals values at each step.

From the obtained Line Chart, we could clearly see the highlighted final data in the chart is the forecasting air passenger number for 15 steps ahead.

Introduction to Time Series Data Analysis and Neural Network – Lab



Data Table - Orange

Info
15 instances (no missing data)
15 features
No target variable.
No meta attributes.

Variables
☒ Show variable labels (if present)
☐ Visualize numeric values
☒ Color by instance classes

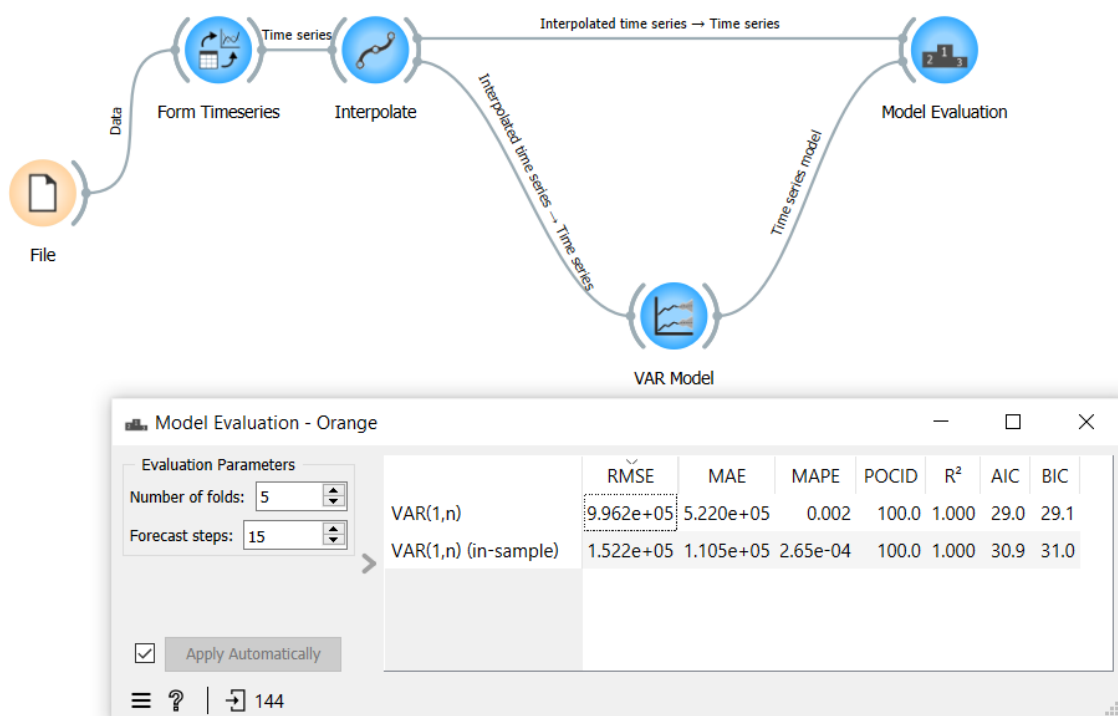
Selection
☒ Select full rows

Restore Original Order
☒ Send Automatically

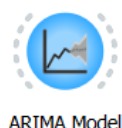
	Passengers (forecast)	Month (forecast)	Passengers (mean) (forecast)	Passengers (min) (forecast)	Passengers (max) (forecast)	Passengers (original)
1	424.076	-2.84031e+08	452.118	376.638	537.341	424.076
2	418.434	-2.81504e+08	440.418	368.945	516.383	418.434
3	413.32	-2.79027e+08	432.545	363.019	505.522	413.32
4	408.416	-2.76585e+08	425.992	357.858	497.314	408.416
5	403.651	-2.74173e+08	420.112	353.134	490.164	403.651
6	398.998	-2.71789e+08	414.664	348.7	483.623	398.998
7	394.439	-2.69431e+08	409.517	344.474	477.493	394.439
8	389.965	-2.67098e+08	404.591	340.402	471.658	389.965
9	385.565	-2.64791e+08	399.83	336.448	466.043	385.565
10	381.235	-2.62507e+08	395.2	332.592	460.598	381.235
11	376.97	-2.60247e+08	390.677	328.815	455.29	376.97
12	372.768	-2.5801e+08	386.245	325.109	450.094	372.768
13	368.627	-2.55795e+08	381.893	321.466	444.997	368.627
14	364.543	-2.53603e+08	377.612	317.88	439.988	364.543
15	360.516	-2.51434e+08	373.399	314.349	435.059	360.516

Evaluate different time series' models. by comparing the errors they make in terms of: root mean squared error (RMSE), median absolute error (MAE), mean absolute percent error (MAPE), prediction of change in direction (POCID), coefficient of determination (R^2), Akaike information criterion (AIC), and Bayesian information criterion (BIC).

- Number of folds for time series cross-validation.
- Number of forecast steps to produce in each fold.
- Results for various error measures and information criteria on cross-validated and in-sample data.



EXPLORATION



By using the ARIMA model, forecast the same dataset. Is the result obtained as same as with VAR model?

PART E: Case Study

 daily-minimum-temperatures-in-me.csv	07-Oct-19 2:43 AM	Microsoft Excel Co...	55 KB
 Electric_Production.csv	07-Oct-19 2:43 AM	Microsoft Excel Co...	8 KB
 sales-of-shampoo-over-a-three-ye.csv	07-Oct-19 2:43 AM	Microsoft Excel Co...	1 KB

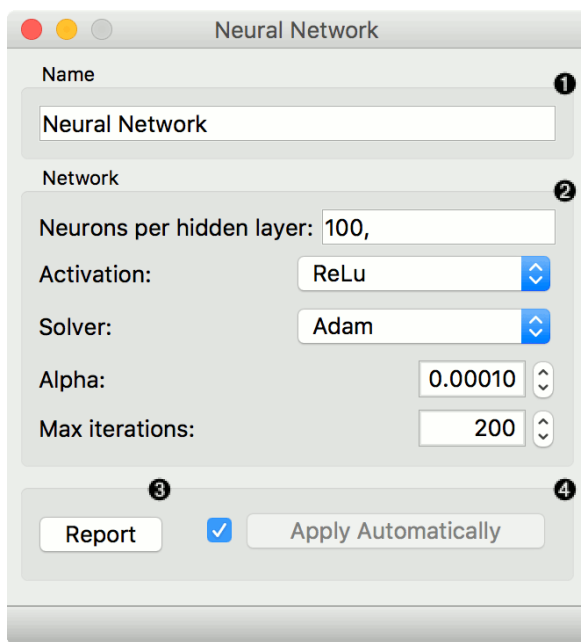
By using the dataset that provided to you, do a data forecasting for 20 steps ahead using VAR and ARIMA model. You will need some data pre-processing techniques in order to do the forecasting.

Working with Neural Network

PART A: Introduction

Orange provided neural network in their widget. You can find the widget on the left panel at “Model” section. Generally, it is also a classifier for supervised learning.

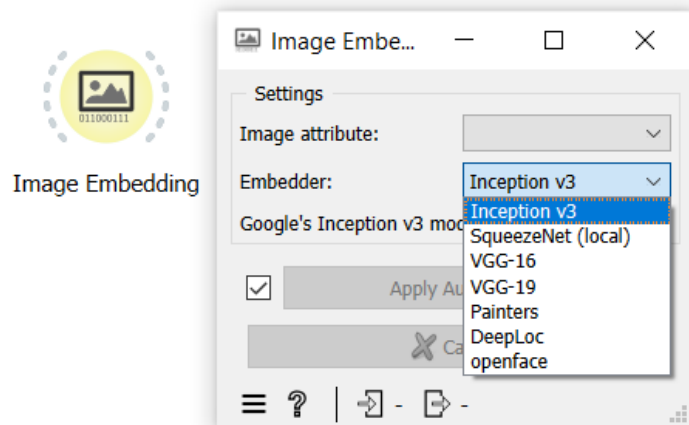
The Neural Network widget uses sklearn's Multi-layer Perceptron algorithm that can learn non-linear models as well as linear.



- A name under which it will appear in other widgets. The default name is "Neural Network".
- Set model parameters:
 - Neurons per hidden layer: defined as the i th element represents the number of neurons in the i th hidden layer. E.g. a neural network with 3 layers can be defined as 2, 3, 2.
 - Activation function for the hidden layer:
 - Identity: no-op activation, useful to implement linear bottleneck
 - Logistic: the logistic sigmoid function
 - tanh: the hyperbolic tan function
 - ReLu: the rectified linear unit function
- Solver for weight optimization:
 - L-BFGS-B: an optimizer in the family of quasi-Newton methods
 - SGD: stochastic gradient descent
 - Adam: stochastic gradient-based optimizer
 - Alpha: L2 penalty (regularization term) parameter
- Max iterations: maximum number of iterations

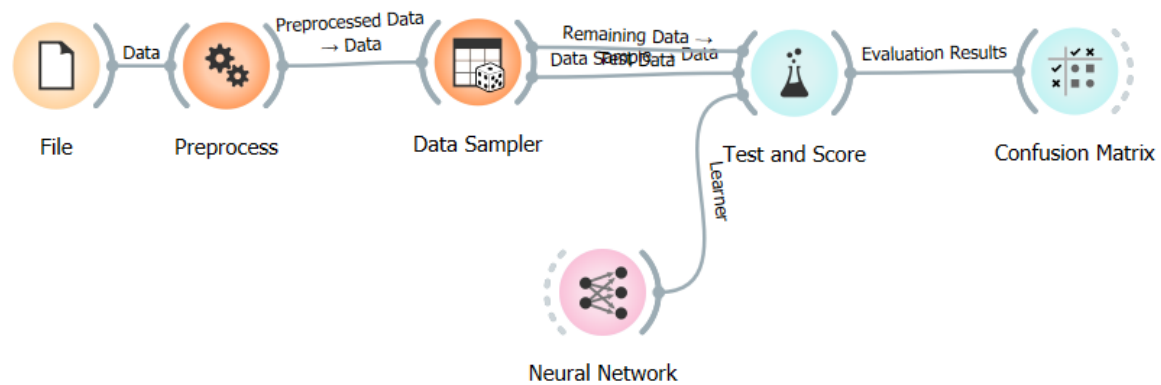
- Other parameters are set to sklearn's defaults.
- Produce a report.
- When the box is ticked (Apply Automatically), the widget will communicate changes automatically. Alternatively, click Apply.

Other NN model such as Convolution Neural Network (CNN) i.e. transfer learning can be found in “Image Analytics” widget.

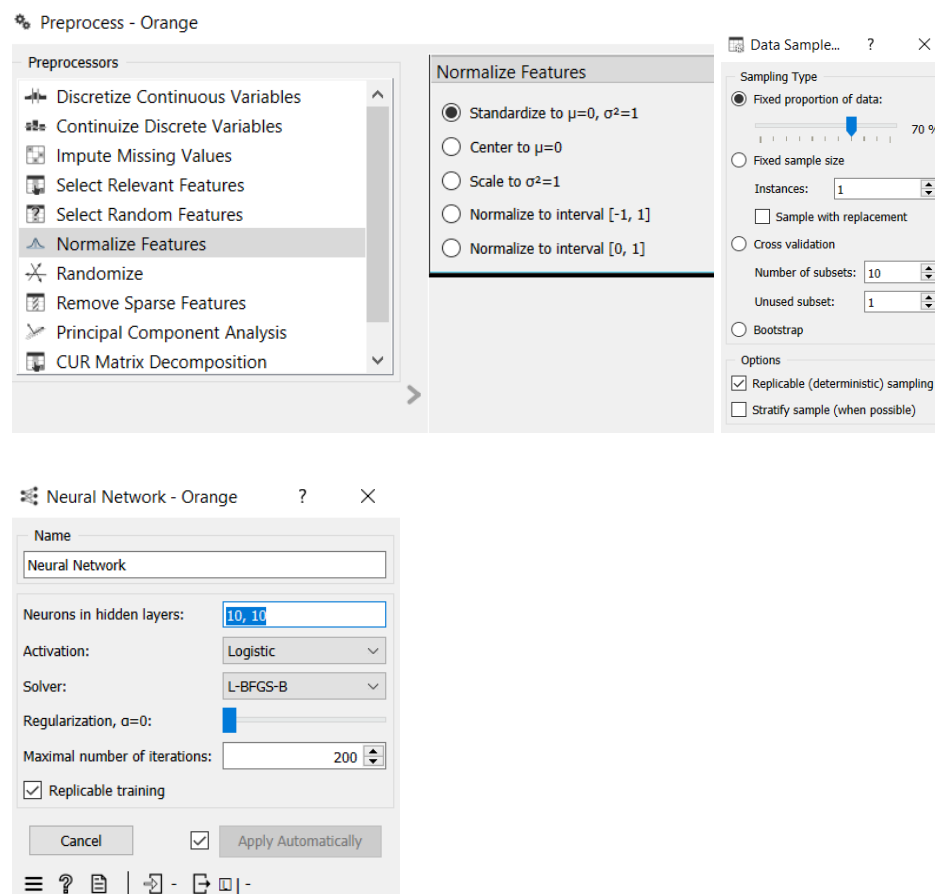


PART B: Classification using NN

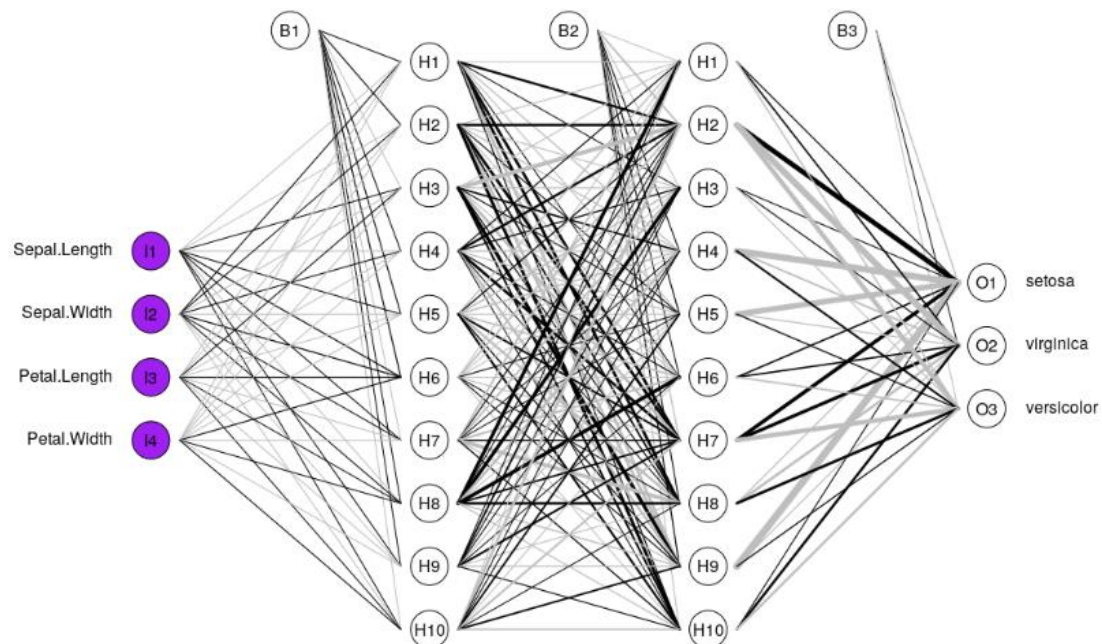
We will use iris.tab dataset to simulate a simple classification using NN. The workflow is as follows:



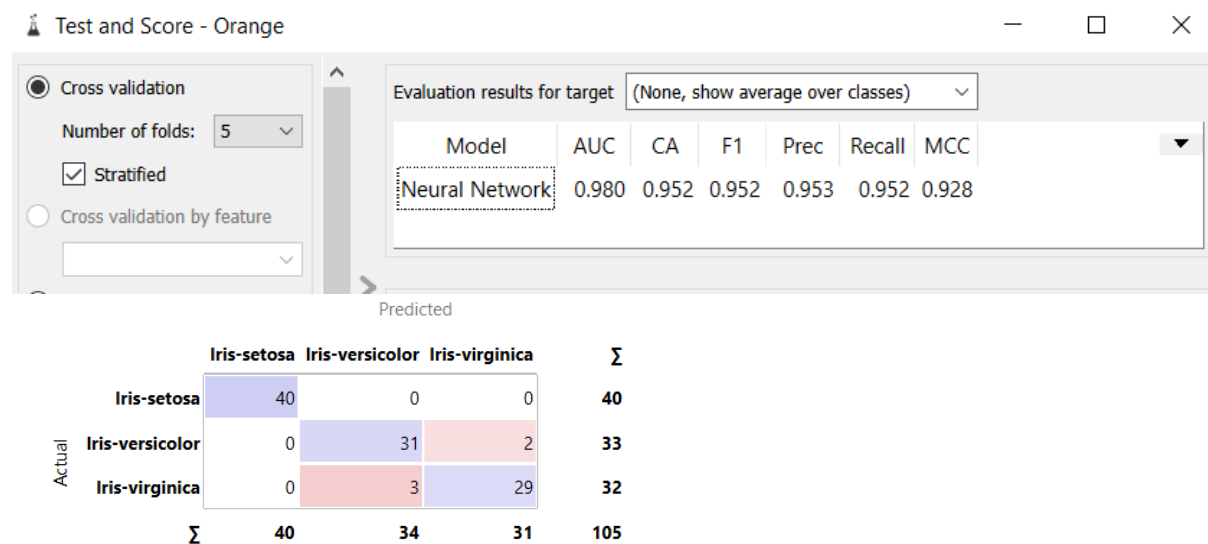
The setup of each workflow are as follows:

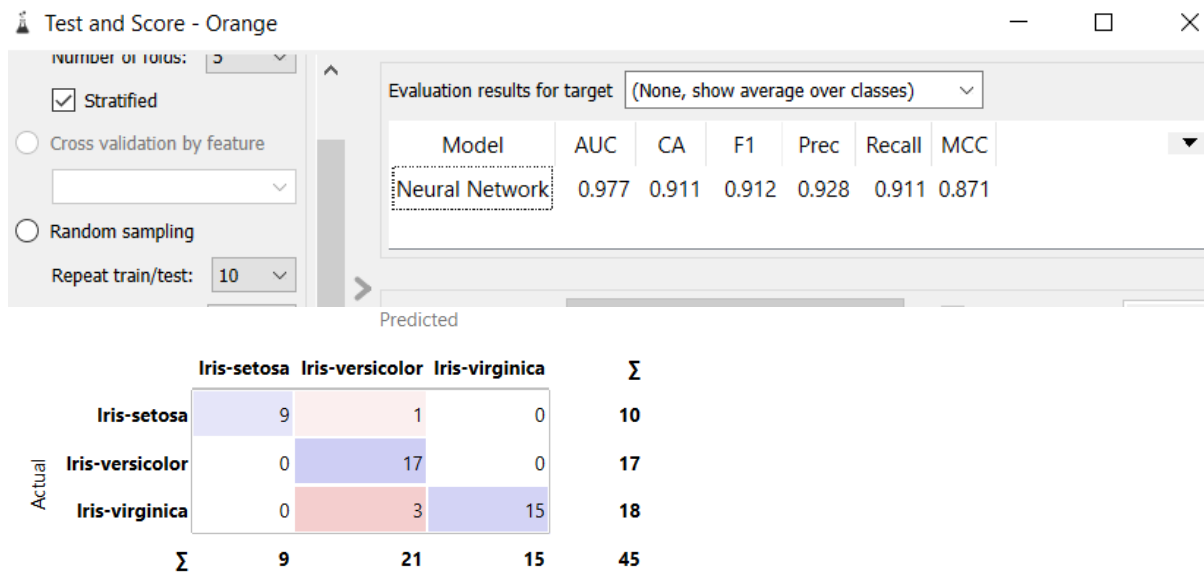


By using the above neural network setup, the visualisation of neural networks are shown in the following figure. There 10 + 10 hidden layers, 4 inputs and 3 outputs. In MLP, all layers are connected to each other in multi dimensions.



The performances of the NN models are:





EXPLORATION

By using the similar dataset, change the setup of your MLP NN. Will the performance results will be the same as previous?

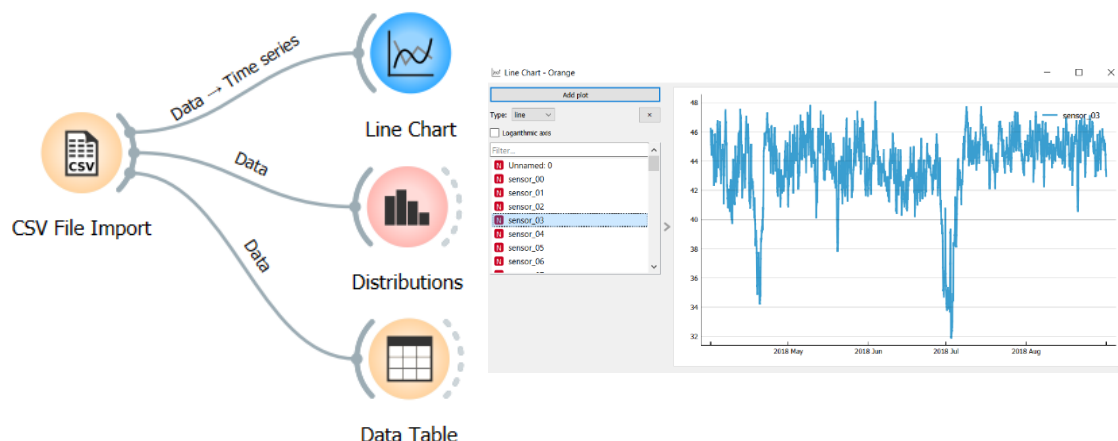
PART C: Understanding time series data for classification

In previous example, we already use a time series data for forecasting the future air passenger. We will use several time series data to understand what kind of dataset is suitable for classification or predictions/forecasting. In terms of meaning, predictions and forecasting is sharing a similar meaning whereby to predict/forecast something in future i.e. forecasting the number of future passenger. Meanwhile, classification is a process to classify something if the obtained parameters are in certain values.

We will use the sensor pump data for this example. You can find the dataset namely “sensor.csv” in the given folder.

The statement of the problem:

I have a friend who working in a small team that taking care of water pump of a small area far from big town, there are 7 system failure in last year. Those failure cause huge problem to many people and lead to some serious living problem of some family. The team can't see any pattern in the data when the system goes down, so they are not sure where to put more attention. Since I believe in using data to solve problem, I ask him to provide available sensor data and hope that someone here can help.



Data Table - Orange

Info
220320 instances
55 features (2.8 % missing data)
No target variable.
No meta attributes.

Variables
☒ Show variable labels (if present)
☐ Visualize numeric values
☒ Color by instance classes
Restore Original Order
☒ Send Automatically

	Unnamed: 0	timestamp	sensor_00	sensor_01
1	28	2018-04-01 00:...	2.45452	47.
2	28	2018-04-01 00:...	2.45452	47.
3	28	2018-04-01 00:...	2.45452	47.3
4	28	2018-04-01 00:...	2.45452	47.
5	28	2018-04-01 00:...	2.45452	47.1
6	28	2018-04-01 00:...	2.45452	47.
7	28	2018-04-01 00:...	2.45452	47.0
8	28	2018-04-01 00:...	2.45452	47.1
9	28	2018-04-01 00:...	2.45452	47.0

220k 1 | 220k

This file contains 3 main group of data.

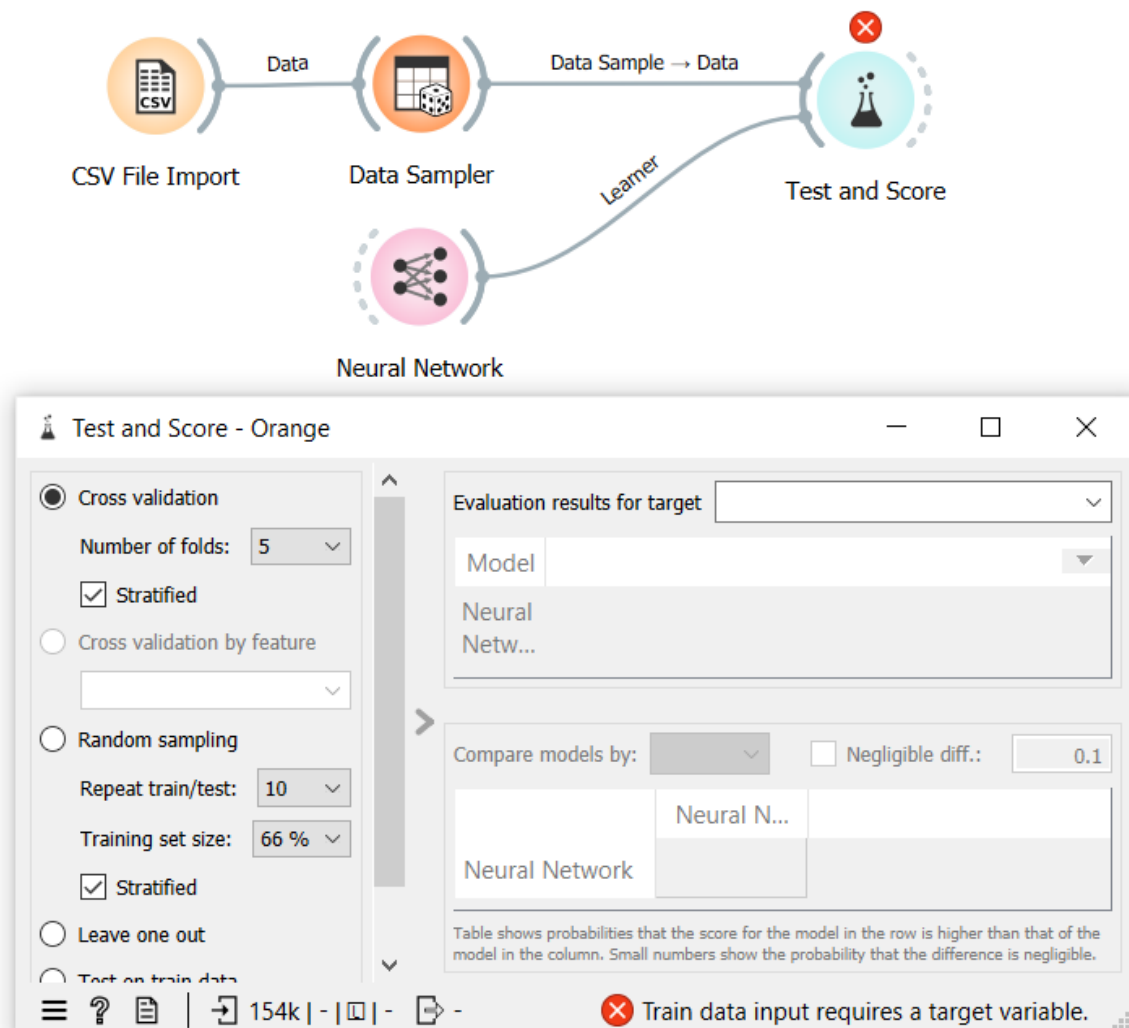
- Timestamp data
- Sensor data (52 series): All values are raw values.
- Machine status: This is target label that I want to classify when the failure will happen

From the tabulated line chart in Orange, we can also see the time series since the data has a time stamp. In this case the data is take for several months in 2018. Based on the problem given, the purpose of this project is for classify what will happen when sensors values give some reading. This is a classification problem! If the purpose of this project is to forecast what will happen to the pump in future for several months ahead, that is a forecasting problem! The main key point between classify and prediction is classification has a label/cluster, meanwhile prediction does not have the label/cluster.

You will understand further once we move to the next sections. We will go through both problems i.e. classification and predictions.

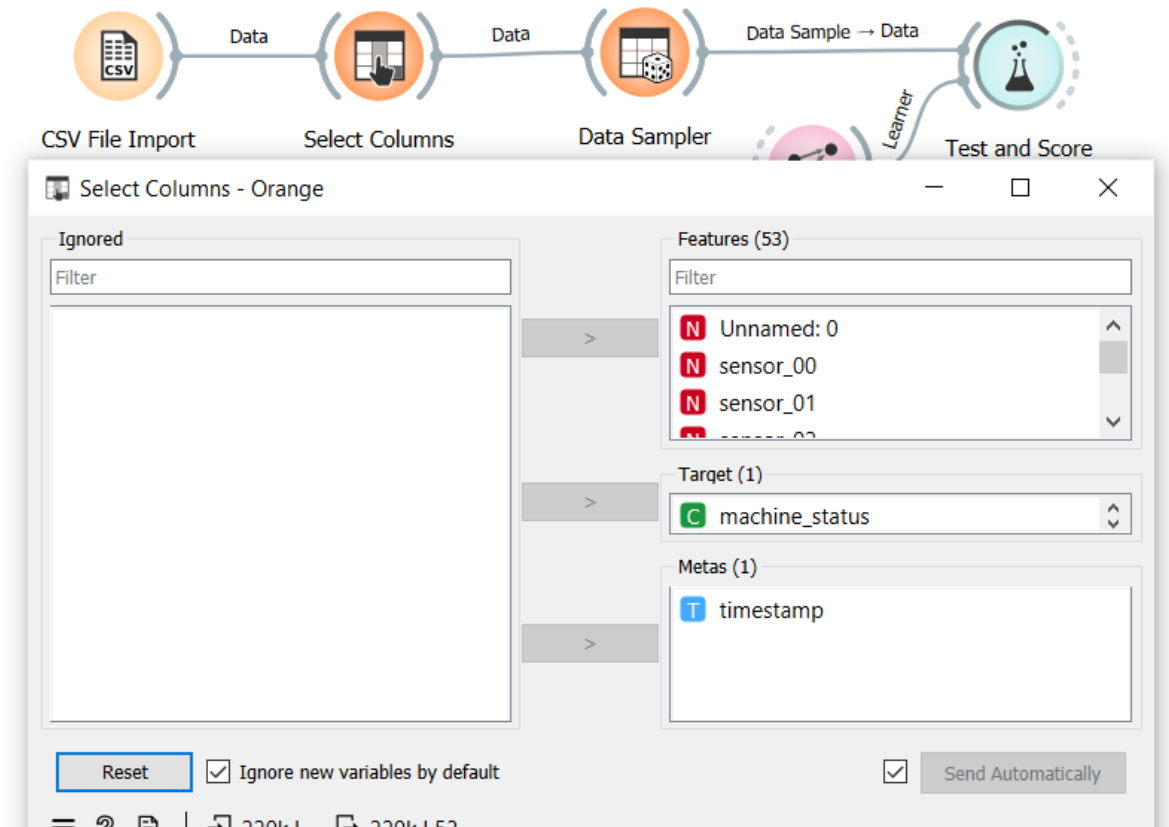
PART D: Time series data classification using NN

To do classification, we use a similar dataset to the previous example “sensor.csv”. Use a similar workflow for classification.



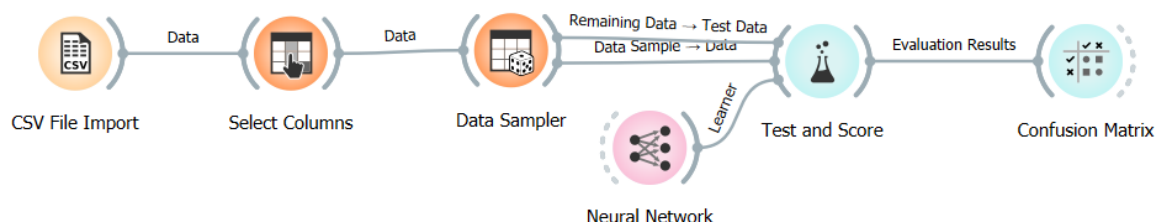
From the observation, we could not train the data since there is no target available in the original dataset. As highlighted before, classification is a process to classify which needs a label/cluster/target.

Select “machine_status” as a target by using Select Columns widget. There are 3 statuses of the machine i.e. Broken, Normal, and Recovery.

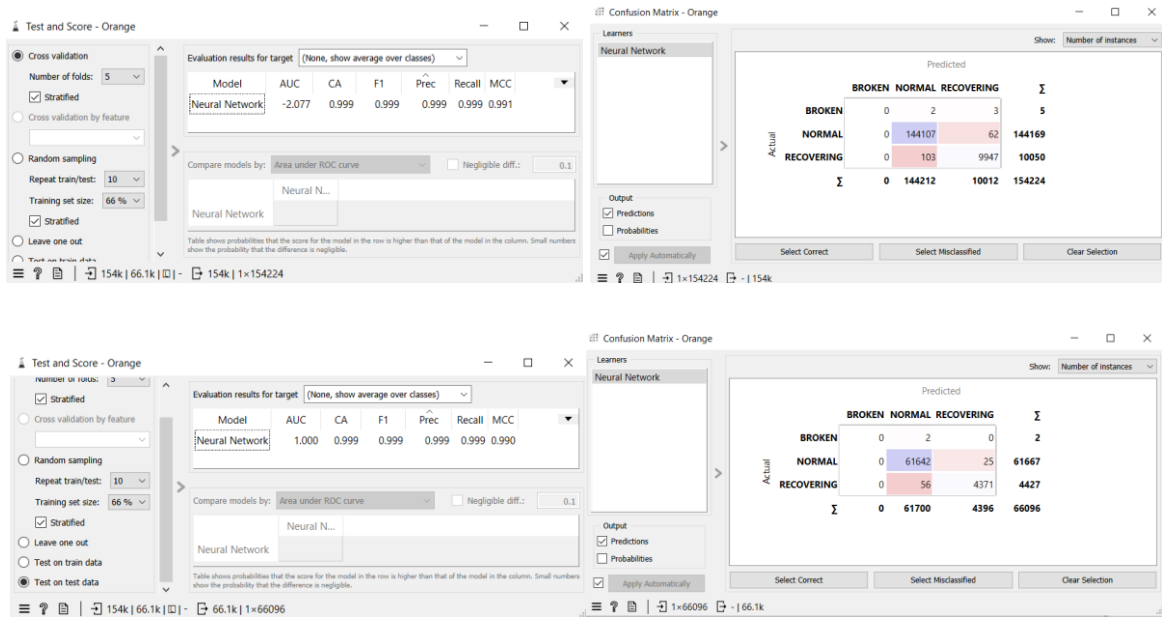


Now there is no more error in the workflow. We can get the results and confusion matrix of the classification. From the observation, we can see that the dataset is imbalanced. The best way to do classification is by having an imbalanced dataset.

The purpose of this project is to classify what will happen to the machine when sensor values give some reading. We imagine that the test sample is unknown and we do not know what will happen to the machine. From the results, we can conclude that only 1% of the model fails to classify what will happen to the machine when sensor values give some reading.



Introduction to Time Series Data Analysis and Neural Network – Lab



EXPLORATION

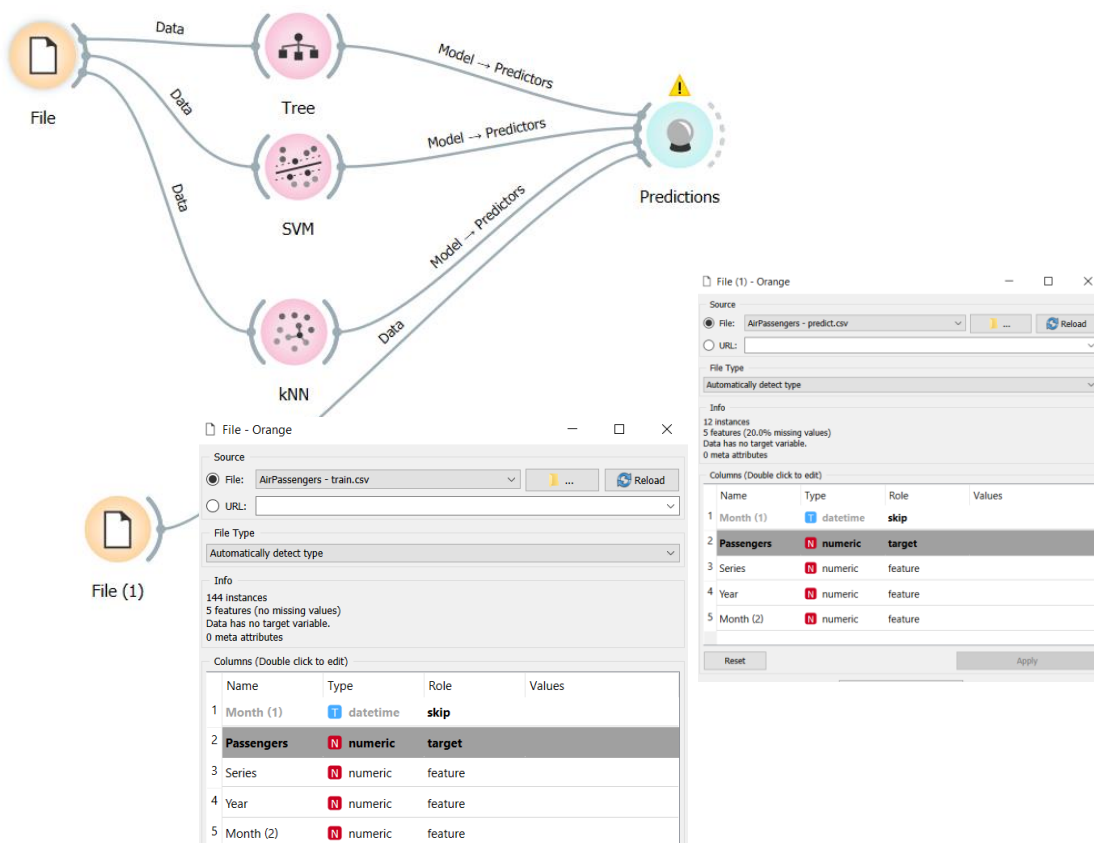
By using other classifiers i.e. SVM, Tree, SVM, kNN, classify the above data. Show the output. Which classifiers give the best result to classify the sensor data?

PART E: Time series data prediction using Machine Learning

We will use a “AirPassengers.csv” data for this example. Let’s say we want to predict the passenger for 12 months in 1961. First of all, we need to do some alteration to the original data. We added three more columns i.e. Series (series number), Year, and Month. Save the altered data into “AirPassenger – train.csv”. This will be our training set. Create one more file namely as “AirPassengers – predict.scv”. This will be our prediction data. We just make an empty column to the #Passenger attribute.

	A	B	C	D	E	F		A	B	C	D	E	F
	Month	#Passenge	Series	Year	Month			Month	#Passenge	Series	Year	Month	
1	1949-01	112	1	1949	1		2	1961-01		145	1961	1	
2	1949-02	118	2	1949	2		3	1961-02		146	1961	2	
3	1949-03	132	3	1949	3		4	1961-03		147	1961	3	
4	1949-04	129	4	1949	4		5	1961-04		148	1961	4	
5	1949-05	121	5	1949	5		6	1961-05		149	1961	5	
6	1949-06	135	6	1949	6		7	1961-06		150	1961	6	
7	1949-07	148	7	1949	7		8	1961-07		151	1961	7	
8	1949-08	148	8	1949	8		9	1961-08		152	1961	8	
9	1949-09	136	9	1949	9		10	1961-09		153	1961	9	
10	1949-10	119	10	1949	10		11	1961-10		154	1961	10	
11	1949-11	117	11	1949	11		12	1961-11		155	1961	11	
12	1949-12	118	12	1949	12		13	1961-12		156	1961	12	
13							14						

After that, setup your workflow as in the following figure. Load “AirPassenger – train.csv” to File widget and “AirPassengers – predict.scv” to File (1) widget. Use a default parameter for all classifiers.



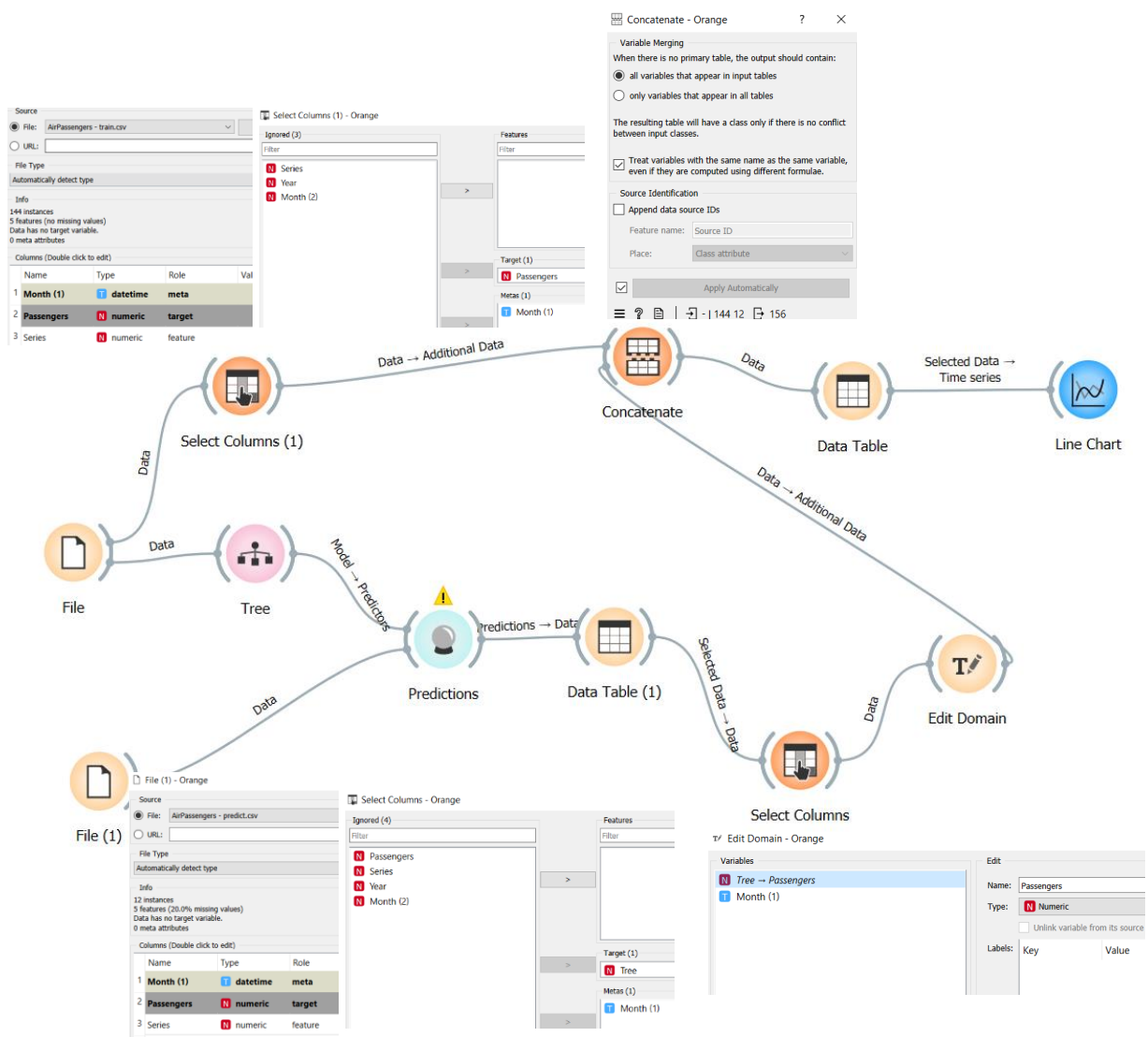
Open Predict widget and you will see the predicted #Passengers using all three classifiers at left panel.

Predictions - Orange

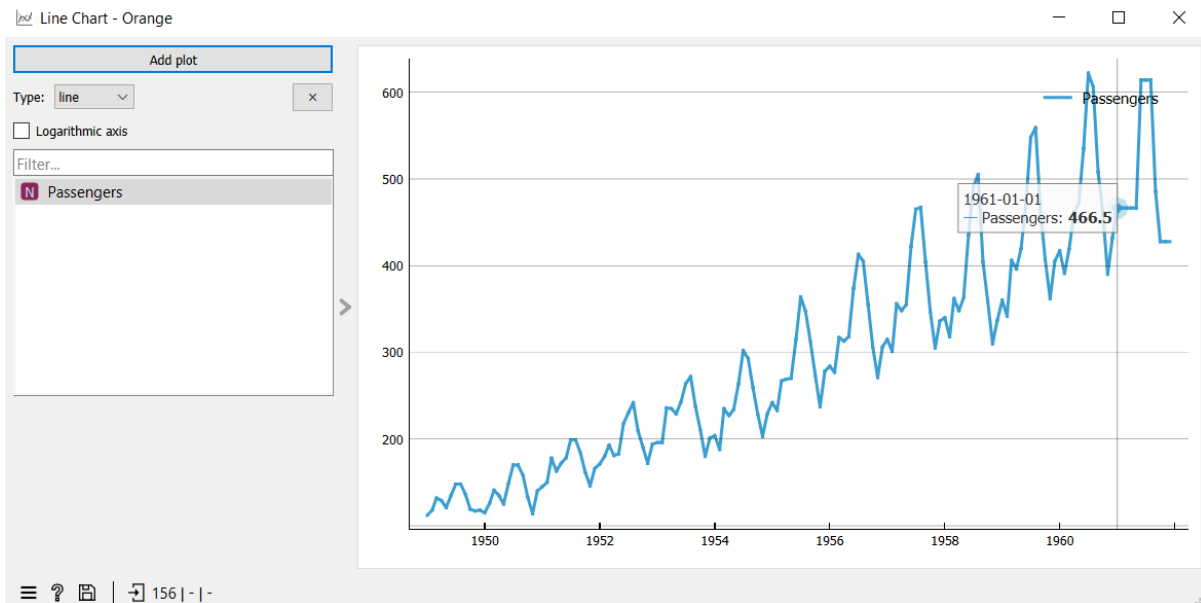
Shown regression error: (None) Restore Original Order

	Tree	SVM	kNN	Passengers	Series	Year	Month (2)
1	466	281	549	?	145	1961	1
2	466	283	546	?	146	1961	2
3	466	286	517	?	147	1961	3
4	466	288	479	?	148	1961	4
5	466	289	479	?	149	1961	5
6	614	290	479	?	150	1961	6
7	614	290	479	?	151	1961	7
8	614	289	479	?	152	1961	8
9	486	287	479	?	153	1961	9
10	428	285	479	?	154	1961	10
11	428	282	479	?	155	1961	11
12	428	280	479	?	156	1961	12

To visualise all predicted value together with the original values, we need to concatenate the original data and the predicted data in one table.



Finally we can the output at Line Chart widget.



From the figure, the new forecast data is started from the 1961-01-01.

EXPLORATION

	daily-minimum-temperatures-in-me.csv	07-Oct-19 2:43 AM	Microsoft Excel Co...	55 KB
	Electric_Production.csv	07-Oct-19 2:43 AM	Microsoft Excel Co...	8 KB
	sales-of-shampoo-over-a-three-ye.csv	07-Oct-19 2:43 AM	Microsoft Excel Co...	1 KB

By using the given data in the folder, do a data forecasting for 20 steps ahead. You will need some data pre-processing techniques in order to do the forecasting.